# Pin-Jun Chen

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## RESEARCH INTERESTS

**Advanced Packaging Design:** enhance the performance of TSV-based/monolithic 2.5D/3D ICs by multiphysics design.
**AI Hardware:** leverage heterogeneous integration to realize scalable, energy-efficient, and high-speed chips for AI/ML.

## EDUCATION BACKGROUND

**Georgia Institute of Technology (Georgia Tech)** (GPA: 3.83/4.00) **Atlanta, GA**
*PhD in Electrical and Computer Engineering* *August 2024 – Expected May 2029*

- **Research Advisor:** Dr. Shimeng Yu, Full Professor, Georgia Tech.
- **Relevant Coursework:** Microelectronics Systems Packaging (A), Hardware Software Co-Design for Machine Learning Systems (A), Photonics for AI (A), Memory Device Technologies (A)

**Georgia Institute of Technology (Georgia Tech)** **Atlanta, GA**
*MSc in Computational Science and Engineering* *August 2026 – Expected December 2027*

- **Relevant Coursework:** Machine Learning (A), Advanced Programming Techniques (A), Computational Data Analysis, Numerical Linear Algebra, Modeling and Simulation, Algorithms.

**National Chiao Tung University (NCTU)** (GPA: 4.07/4.30) **Hsinchu, Taiwan**
*MSc in Semiconductor Technology* *September 2017 – May 2020*

- **Research Advisor:** Dr. Kuan-Neng Chen, Chair Professor, NCTU
- **Master's Thesis: "***Process Development and Simulation Analysis for Lowering Thermal Budget on Stacked/Monolithic 3D Integrated Circuits"*
- **Relevant Coursework:** Semiconductor Processing (A+), Semiconductor Physics and Devices (A), Nanometer Scale Electronic Devices (A), Reliability and Failure Physics of Semiconductor Devices (A-), 3D Integrated Circuit (A-)

**National Tsing Hua University (NTHU)** (GPA: 3.58/4.30) **Hsinchu, Taiwan**
*BSc in Materials Science and Engineering* *September 2013 – June 2017*

- **Relevant Coursework:** Electromagnetism (A+), Engineering Mathematics (A+), Electronics Laboratory (A+), Calculus (A), The Physical Properties of Materials (A), Introduction to Quantum Physics (A-)

## PUBLICATIONS & CONFERENCE PRESENTATIONS (Google Scholar)

- Wei-Hsing Huang, Janak Sharda, Cheng-Jhih Shih, Yuyao Kong, Faaiq Waqar, **Pin-Jun Chen**, Yingyan (Celine) Lin, and Shimeng Yu, "A3D-MoE: Acceleration of Large Language Models with Mixture of Experts via 3D Heterogeneous Integration," *IEEE Journal on Exploratory Solid-State Computational Devices and Circuits*, July 2026, doi: 10.1109/JXCDC.2026.3713957
- **Pin-Jun Chen**, Janak Sharda, Po-Kai Hsu, Min Gyu Park, Sung Kyu Lim, Muhannad Bakir, and Shimeng Yu, "3D Stacked GPU Architecture with Ultra-Thin 3D DRAM Dies and Voltage Regulators Embedded into Glass Substrate," *in IEEE 76th Electronic Components and Technology Conference (ECTC)*, Orlando, FL, May 26-29, 2026, doi: 10.1109/ECTC51846.2026.00241. **(oral presentation)**
- Tzu-Chieh Chou, Shin-Yi Huang, **Pin-Jun Chen**, Han-Wen Hu, Demin Liu, Chih-Wei Chang, Tzu-Hsuan Ni, Chao-Jung Chen, Yu-Min Lin, Tao-Chih Chang, and Kuan-Neng Chen, "Electrical and reliability investigation of Cu-to-Cu bonding with silver passivation layer in 3D integration," *IEEE Transactions on Components, Packaging and Manufacturing Technology (TCPMT)*, vol. 11, no. 1, pp. 36-42, Nov. 2020, doi: 10.1109/TCPMT.2020.3037365.
- Ping-Yi Hsieh, Yi-Jui Chang, **Pin-Jun Chen**, Chun-Liang Chen, Chih-Chao Yang, Po-Tsang Huang, Yi-Jing Chen, Chih-Ming Shen, Yu-Wei Liu, Chien-Chi Huang, Ming-Chi Tai, Wei-Chung Lo, Chang-Hong Shen, Jia-Min Shieh, Da-Chiang Chang, Kuan-Neng Chen, Wen-Kuan Yeh, and Chenming Hu, "Monolithic 3D BEOL FinFET switch arrays using the location-controlled-grain technique in voltage regulator with better FOM than 2D regulators," in *IEEE 65th International Electron Devices Meeting (IEDM)*, San Francisco, CA, Dec. 7-11, 2019, pp. 3.1.1-3.1.4, doi: 10.1109/IEDM19573.2019.8993441.
- **Pin-Jun Chen**, Chih-Ming Shen, Chih-Chao Yang, Ming-Chi Tai, Wei-Chung Lo, Chang-Hong Shen, Chenming Hu, and Kuan-Neng Chen ,"Transient thermal damage simulation for novel location-controlled grain technique in monolithic 3D IC," in *IEEE 14th International Microsystems, Packaging, Assembly and Circuits Technology Conference (IMPACT)*, Taipei, Taiwan, Oct. 23-25, 2019, pp. 104-107, doi: 10.1109/IMPACT47228.2019.9024960. **(oral presentation)**

## RESEARCH EXPERIENCE

**Georgia Institute of Technology** **Atlanta, GA**
*Graduate Research Assistant, School of Electrical and Computer Engineering* *August 2024 – Present*
- Design advanced 3D architectures to improve power, performance, and area (PPA) parameters for AI accelerators.
- Conducted multiphysics analysis and device simulation to characterize emerging memory technologies for AI/HPC applications.

**Katholieke Universiteit Leuven (KU Leuven) / imec** **Leuven, Belgium**
*International Scholar / R&D Intern, Wafer Bonding and Assembly* *January 2020 – May 2020*
- Granted admission as an International Scholar in the Faculty of Science at KU Leuven.
- Operated a laminator, bonder, and SAM to explore the optimal parameters for the dry film photoresist (DFR) process.
- Fabricated 3D microfluidic devices using the DFR technique with novel processes implemented.

**NCTU-TSMC Joint Research Center** **Hsinchu, Taiwan**
*Research Assistant, Center for Advanced Semiconductor Technology Research* *September 2017 – September 2019*
- Developed the next-generation monolithic back-end-of-line 3D FinFETs in collaboration with Taiwan Semiconductor Research Institute and NCTU-Berkeley Joint Research Center.
- Designed the laboratory's first thermal simulation environment for monolithic 3D ICs.
- Utilized ANSYS to analyze transient thermal conduction in various FinFET models under different boundary conditions.

**Industrial Technology Research Institute** **Hsinchu, Taiwan**
*Research Intern, Electronics and Optoelectronics Research Laboratories* *May 2018 – April 2019*
- Employed various materials as passivation layers to improve the Cu-to-Cu bonding technique.
- Enhanced the manufacturability of TSV-based 3D stacked ICs by reducing the thermal budget in processes.

**The Chinese University of Hong Kong (CUHK)** **Hong Kong, China**
*Undergraduate Research Assistant, Department of Chemistry* *July 2017 – August 2017*
- Research topic: Synthesis of Sonogashira Cross-Coupling Reaction in Prof. Chit Tsui’s lab.
- Used Pd catalyst to synthesize over 20+ types of compounds to investigate the Sonogashira coupling mechanism.

## INDUSTRY EXPERIENCE

**Ericsson** **Taipei, Taiwan**
*5G Early Career Program Trainee, Market Area Northeast Asia* *November 2020 – March 2023*
- Selected as one of the seven elite trainees from a diverse pool of candidates across Taiwan, China, Hong Kong, Macau, Japan, and South Korea, to advance regional 5G initiatives.
- Promoted Ericsson’s cloud radio access network in IEEE GLOBECOM 2020, one of the IEEE Communications Society’s flagship conferences dedicated to driving innovation in nearly every aspect of communications.
- Led and managed the projects, provided solutions for Asia Pacific Telecom and Far EasTone Telecom in 4G/5G network deployment (LTE 700M/LTE 2.6G, NR 28G), establishing Taiwan's first 5G non-standalone multi-operator core network for increased throughput and improved subscriber experiences, particularly in rural and mountainous areas.

**Qualcomm** **Hsinchu, Taiwan**
*Hardware Engineering Intern, Reliability Engineering* *May 2019 – November 2019*
- Acquired knowledge in product-level reliability evaluation for cutting-edge silicon processes used in mobile products.
- Learned to use statistical analysis techniques such as Weibull and Lognormal distributions, reliability tools or software, high-temperature-operating-life ovens, and electrostatic discharge machines.

## HONOURS & AWARDS

- **Taiwan Government Scholarship for Study Abroad**, Ministry of Education, Taiwan — 2026
  Nationally competitive three-year scholarship supporting doctoral study abroad.
- **Indigenous Peoples Scholarship for Overseas Study**, Council of Indigenous Peoples, Taiwan — 2025
  Competitive scholarship supporting overseas graduate study.
- **KU Leuven Scholarship**, KU Leuven — 2020
  One-year full scholarship supporting study and research at KU Leuven.
- **Future Tech Breakthrough Award**, Ministry of Science and Technology, Taiwan — 2019
  Joint research project selected as a winner of the national Future Tech Breakthrough Award.

- **Graduate Research Assistant Excellence Award**, TSMC–NCTU Joint Research Center — 2019
  Awarded in recognition of outstanding research performance.
- **Helm Technology Scholarship**, National Chiao Tung University — 2018, 2019
  Awarded to one student from each college for exceptional academic performance.
- **2017 CUHK Summer Exchange Scholarship**, The University System of Taiwan — 2017

## TECHNICAL SKILLS & LANGUAGES

- **Integrated Circuits Fabrication:** Photolithography, Sputtering, Wafer Bonding, Dry Film Lamination
- **Material Analysis:** Scanning Electron Microscopy, Energy Dispersive Spectroscopy, Scanning Acoustic Microscopy
- **Simulation Toolkits:** ANSYS Workbench/HFSS/Fluent/Lumerical, Synopsys TCAD, Candence Spectre/Innovus/Virtuoso, Simens xSI/xPD/HyperLynx, Keysight ADS
- **Programming:** C/C++, Python, MATLAB
- **Languages:** Mandarin (Native), Taiwanese (Native), Hakka (Native), and English (Proficient)